Independent Equipment Reliability & Operating Envelope Review

Woodmizer MR6000 Dual Arbor Gang Saw

Tier 1 Deliverable – Reliability Engineering Audit (Client-Facing Case Study)

Prepared by KLB Industrial Corp.

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1. Executive Summary

A Tier 1 reliability audit and corrective action package was executed on a dual arbor gang saw system equipped with 222-series spherical roller bearings mounted on tapered adapter sleeves. The system is driven by two 200 HP electric motors via 5VX belt transmission with arbor speeds typically in the 1800–2200 RPM range. The work scope prioritized bearing temperature control, precision alignment, contamination control, and repeatable installation quality. The primary reliability lever was the replacement of an extremely high-viscosity grease (ISO VG 1500) with a speed-appropriate EP grease specification, paired with an installation SOP that controls internal clearance reduction, prevents brinelling, and enforces cleanliness.

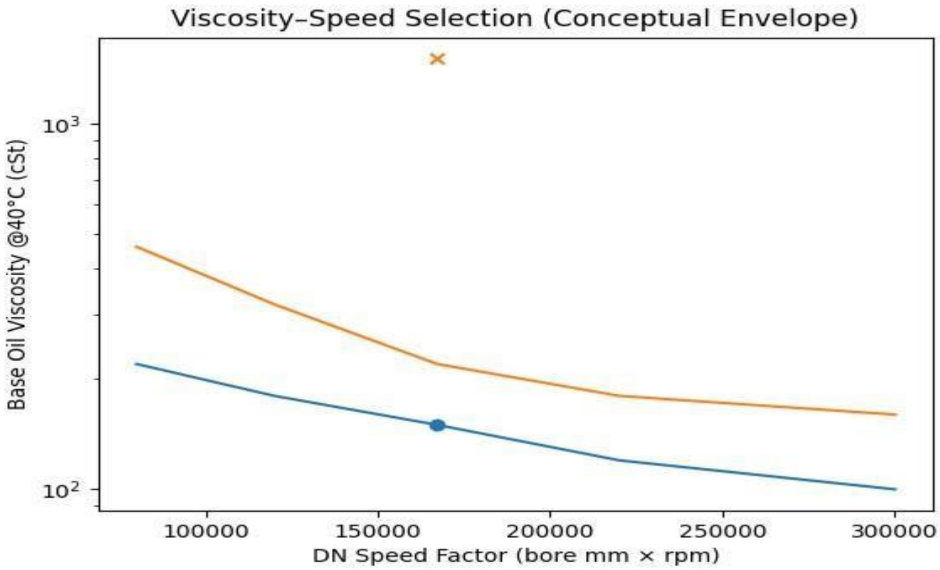
Key Engineering Snapshot

Parameter	Value
Arbor speed envelope	1800–2200 RPM (sub-3000 RPM capability)
Shaft diameter (approx.)	3.0 in (76 mm)
Bearing family	222- series spherical roller bearings (tapered bore on adapter sleeve)
DN speed factor (approx.)	167,200
Drive type	Belt-driven (5VX profile)

2. Lubrication Engineering – Grease Selection &Justification

Observed Condition: The bearing system was lubricated with an ISO VG 1500 synthetic base oil grease. Greases in the ISO VG 1500 class are intended for *extremely slow speed*, heavy-load, boundary-lubrication regimes (e.g., kiln rollers, severe roll press applications). In higher-speed rolling element bearings, excessive base oil viscosity increases churning losses, elevates bulk grease temperature, and can impair effective oil bleed to the contact zone, particularly when housings are overfilled or purge paths are restricted. The resulting thermal rise accelerates oxidation, reduces grease life, and elevates the probability of premature bearing distress in the presence of alignment or contamination sensitivities.

Engineering Basis for Change: For a DN ≈ 167,000 typical of this application, rolling bearing guidance from major manufacturers supports base oil viscosities broadly in the ISO VG 100–320 range depending on load, temperature, and bearing size. The selected target is an EP grease with lithium complex thickener, NLGI 2 consistency, and ISO VG ~220 base oil viscosity to balance speed capability with belt-driven radial loading. This viscosity class reduces churning heat versus ISO VG 1500 while maintaining film strength for spherical roller bearing contacts under intermittent load.



Note: chart is a conceptual envelope used for selection logic; final lubricant selection is validated against the bearing manufacturer's speed factor guidance and site temperature conditions.

Approved Grease Specification (Tier 1 Standard)	Requirement
Thickener	Lithium complex (EP/AW)
Consistency	NLGI 2
Base oil viscosity	ISO VG 220 (target); ISO VG 100–150 permitted where speed/temperature permit Verify purge path open; stabilize temperature before re-grease adjustment
Additives	EP/AW, oxidation inhibitor, corrosion inhibitor
Fill control	Housing free volume 25–35% (do not pack housing solid)
Startup practice	

Appendix A – Bearing Installation SOP (222-Series SRB on Tapered Adapter Sleeve)

This SOP is written for spherical roller bearings with tapered bore mounted on adapter sleeves. It is structured to control **cleanliness**, **fit**, **internal clearance reduction**, and **repeatability** using manufacturer-recommended measurement methods (feeler gauge clearance reduction).

A1. Safety & Work Controls (Mandatory)

• Establish lockout/tagout and verify zero energy state.
• Confirm guarding removal plan and re-installation plan.
• Use clean, dedicated bearing installation tools only (spanners, feeler gauges, gloves).
• Do not use hammer-and-drift on locknuts (risk of damage and contamination chips).

A2. Cleanliness Standard (Mandatory)

• Bearings remain sealed in original packaging until immediately before installation.
• Work area cleaned; no grinding, air-hose blowdown, or abrasive work within the bearing zone.
• Wear nitrile gloves; do not touch raceways/rollers with bare hands.
• Clean shaft and housing with lint-free wipes; solvent used only if fully evaporated before assembly.
• Inspect seal lands and shaft for scoring; any groove at seal lip contact is addressed prior to assembly.

A3. Pre-Installation Inspection (Required)

• Verify bearing designation (222-series) and internal clearance class (C2 where specified).
• Verify adapter sleeve, locknut, and locking washer are correct for the shaft diameter.
• Inspect shaft seating for burrs, corrosion, raised metal, and taper damage; remove with fine stone/emery andreclean.
• Verify housing condition, dowel/shim integrity, and fastener condition.
• Confirm purge/relief path in housings is open (prevents overpressure and churning heat).

A4. Grease Packing & Lubrication Prep (Required)

- Pack bearing with approved grease prior to mounting: work grease through roller/cage until it appears on theopposite side.
- Do not rely solely on grease fitting injection to fill the bearing; this can leave voids and dry zones on startup.
- Housing cavity fill is controlled to 25–35% free volume (avoid churning/heat).
- Seals installed and verified prior to startup; damaged seals are replaced (no exceptions).

A5. Adapter Sleeve Positioning (Required)

- Position adapter sleeve on the shaft to the required bearing centerline, threads outboard.
- A screwdriver may be inserted into the sleeve slit to slightly open the sleeve for positioning (do not deform thesleeve).
- Light oil film on sleeve OD may be used to ease bearing sliding (avoid oil on taper seat surfaces that would affectfriction control).

A6. Initial Clearance Measurement – Unmounted Clearance (Required)

- Rotate the bearing rings to center the roller complement.
- Measure radial internal clearance using feeler gauges between the outer ring and the most unloaded roller.• For large bearings or where ring deformation affects readings, determine “true” clearance using multi-point method:

Measure at 3 o’clock (b), 9 o’clock (a), and 12 o’clock (c) without moving the bearing and calculate: 0.5(a + b + c).

- Record initial clearance (baseline) prior to drive-up.

A7. Drive-Up / Clearance Reduction – Mounting on Sleeve (Critical Control Step)

- Install bearing onto sleeve; install lockwasher and locknut.
- Tighten locknut incrementally using proper spanner tool to drive the bearing up the taper.
- After each increment, re-measure clearance reduction using feeler gauge method until target reduction isachieved.
- Target mounted reduction for this application: 0.003–0.004 in (C2 control target).
- Maintain approximately equal clearance on both roller rows; adjust bearing position if required.

A8. Shaft Coupling for True Clearance Readings (Application-Specific Requirement)

- Where shaft train separation or axial float can bias readings, mechanically draw shafts together before finalclearance confirmation.
- Use chain/come-along/lever hoist with **protective padding** at contact points to prevent shaft marring.
- Take final clearance readings only after shafts are drawn together and stabilized.

A9. Prohibitions – Preventing Brinelling & Premature Failure (Mandatory)

- Do not set clearance with belt tension applied or with bearing under radial load.
- Do not hammer locknut while the bearing is loaded; driving up a loaded bearing can cause brinelling (permanentindentation) and race damage.
- Do not rotate over a pinched feeler gauge blade; slide the blade through the clearance with slight resistance.

A10. Locking, Final Assembly & Verification (Required)

- Install locking washer correctly with inner tab in sleeve groove; tighten locknut to seated position without furtherdrive-up.

- Bend the nearest locking washer tang into the locknut slot (do not loosen the nut to align; tighten forward ifneeded).
- Verify shaft runout/alignment after mounting, then re-check temperature stabilization after startup.
- Record: initial clearance, final mounted clearance, nut position/angle marks, lubrication volume, and startuptemperatures.

References (Manufacturer & Industry Guidance)

1) SKF Bearing Maintenance Handbook – mounting and measuring radial internal clearance with feelergauges; multi-point true clearance method (3, 9, and 12 o’clock measurements and 0.5(a+b+c) calculation).

2) NSK Handling Instructions for Bearings – mounting bearings with tapered bores; clearance reductionand confirmation against tables; recommendation to measure reduction in clearance during mounting on tapered shafts/sleeves.

3) Mobilith SHC 1500 product description – ISO VG 1500 grease intended for extremely slow speeds,heavy loads and high temperatures (examples include kiln roller bearings).

4) SKF bearing installation guidance – cautions on feeler gauge measurement and handling duringclearance checks.